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Quantifying Deep Learning based Image Forgery Detection System: A Survey

T.Bharathi
Research Scholar, Department of Computer Science and Engineering
Jawaharlal Nehru Technological University, Anantapur
bharathi342@gmail.com

Dr. V Lokeswara Reddy
Professor, Department of Computer science and Engineering
KSRM College of Engineering
vlreddy74@gmail.com

***Abstract*-The last few years have witnessed a high-pace rise in digital image forgery cases, where the forgers intend to manipulate images or videos to change information or even harass individual(s) for different reasons. On the other hand, the rising internet technologies and enabled applications like social media, e-commerce, and business communication which often use images or videos to circulate information have broadened the horizon for forgers to manipulate images. There are various types of image forgery, such as copy-move forgery, image-splicing, and resampling etc., which involve diverse patterns of manipulation and cause pixel-level disturbances that are challenging to detect, especially with the naked eye. Moreover, the advanced manipulation tools make it challenging to detect and classify image forgery cases. Localization is even more challenging task. Unlike traditional block-based or key-points based image forgery detection methods, which often makes use of the hand-crafted features to detect and localize forgery, the deep learning-based approaches can perform superior. The ability to learn local as well as contextual information from the input images make deep learning models more effective towards image forgery detection and classification. The improved deep models with their feature discrimination power strengthens to yield more accurate and reliable forgery detection and localization. This paper covers various systems for detecting and classifying image forgeries using deep learning. The different deep learning techniques such as the convolutional neural networks (CNN), recurrent neural networks (RNNs), auto-encoders, and other enhanced transfer learning methods such as AlexNet, MobileNet, ResNet, etc., designed towards copy-move forgery detection, splicing detection, and resampling detection etc. are discussed, with their strengths as well as limitations. Additionally, the different datasets available and their specifications are also discussed in details. The different challenges and allied scopes for improvement are also given in this manuscript. The concluding inferences along with the future scopes are also discussed in detail which can help researchers innovate and contribute superior towards image security and attack-resilience tasks.**

Keywords- *Image Forgery Detection, Deep Learning, Copy-Move Forgery Detection, Image Splicing, Resampling.*

I. INTRODUCTION

In recent years, there has been a remarkable surge in affordable hardware, vision computing, and decentralized computing technologies, benefiting various sectors such as forensics [1], surveillance systems, smart manufacturing, smart cities, healthcare, defense, e-commerce, social media, and diverse business communication needs. These applications often employ image data to perform analytics and real-time decisions. Despite such upsurging significances, ironically the last few years have witnessed numerous events of image forgery where the perpetrator causes different kinds of manipulations over the digital images to misuse it for the different ill-intended purposes [1].

Image tampering represents the kinds of image forgery that substitutes certain section(s) within an image with some targeted content. Typically, there are two key image manipulation detection approaches; active methods and passive methods. In active forgery methods, the manipulators often append or embed some additional details such as the digital watermark during image acquisition or later. Though, such manipulation can be possible by the authorized persons as well. Passive methods, unlike active methods, do not depend on additional information to detect forgery. For this reason, they are also referred to as blind methods. Functionally, the passive methods retrieve the spatio-textural features from the digital images and make use of them towards forgery detection. The passive methods can broadly be categorized into the different types such as copy-move, resampling, morphing etc. [2]. A snippet of these attack models is given as follows:

A. Copy-Move Forgery

This image manipulation or forgery method is performed by copying one or more components of a digital image and pasting them in the same image, but at the different spatial locations. These forgeries or image manipulations are usually applied to hide significant sections or components of an image. It is also executed to duplicate certain target object or person that eventually changes the semantic information of the intended image. Fig. 2(a) shows the copy-move forgery cases.

B. Image Splicing

The image splicing (forgery) is same as that of the copy-move forgery, yet, the only difference is that the manipulated or pasted component (or object) are removed from one or more digital images. Image splicing can be performed especially to hide certain target component or image content. It is also done to show certain fake conditions to manipulate semantic information. Fig. 2(b) shows an illustration of image splicing.

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C. Image Inpainting

The image inpainting attack represents an image manipulation case that embodies filling certain section or a "hole" within an input digital image with plausible content. Image inpainting method is usually applied to restore damaged components in digital images. Yet, it can be employed by intruders as well to hide target information from easy visualization. It can also be performed to remove or replace watermark. In real-world conditions, the filled regions can be either copied from some other section of the image or can be synthesized by artificial intelligence tools such as the generative adversarial networks (GAN). However, when a specific section of content is copied and pasted within an image, it can be identified as a case of copy-move forgery.

D. Resampling

In this image manipulation case, the input image is manipulated in such manner that certain object within an input image is resampled (ex. Resized, stretched etc.) to yield convincing composite representation. To achieve a visually convincing image composition, resampling methods are applied in which the different tasks such as image resize, rotation, sectional rotation or stretching etc. are performed. For instance, while generating a composite representation of two persons, one person might have to be resized, stretched, or rotated so as to gain relative heights, target presentation etc. This mechanism is called resampling in which the original image is resampled to yield an updated image lattice. However, such resampled images give rise to certain periodic correlations amongst the adjoining pixels. Since, such correlations are impractical to happen (in natural cases) their presence can be quantified to detect manipulation.

E. DeepFakes

In the last few years, DeepFake has emerged as a decisive type of image manipulation or forgery case, in which the perpetrator often makes use of the deep learning methods such as GAN to synthesize fake contents within an input image or video. Before the deep-era, such manipulations were done manually by applying certain professional editing software. Though, the development in deep technologies such as the GAN technology has broadened its horizon where one can generate misleading video or images even with minimum perceptibility. Contemporarily, the perpetrators use DeepFake tools to replace one's face and map target face with another person, intentionally done to extort, socially defame and blackmail.

Unlike other image manipulation cases, the identification of the forgery in copy-move and splicing cases is really challenging. Unlike active methods, image forgery detection methods like watermarking, passive methods are of greater significance; yet, their computational challenges and complexities can't be ruled out. There are a number of evidences in which the active methods insert watermarks and other digital signatures to the digital images to ensure (image) authenticity. The passive methods on the other hand enable alters the detail(s) to manipulate the image data and overshadow vital images. Figure 2 illustrates various types of image forgery. Figure 2(a) depicts examples of copy-move forgery, where (a) shows the original image and (b) shows the forged (copy-move tampered) image. In contrast, Figure 2(b) depicts cases of image splicing, where the first and second images are the originals, and the spliced image is shown on the right side of Figure 2(b). The illustration of image morphing is given in Fig. 2(c), where the initial two images are the original images, while the morphed one is given in the right.

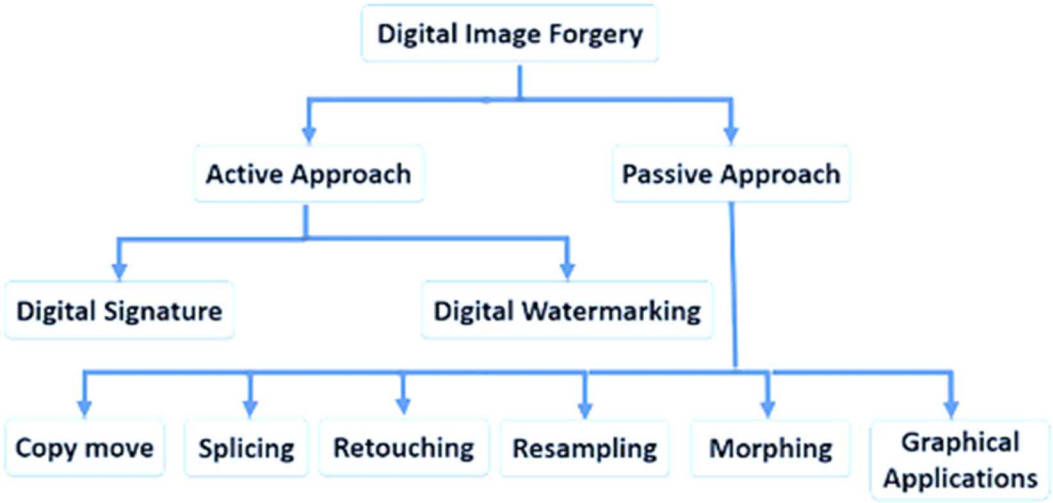

Fig. 1 Image forgery detection techniques

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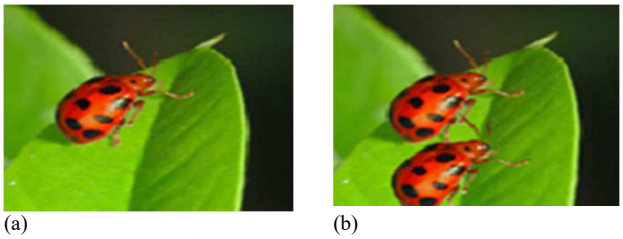

(a) (b)

Fig. 2(a) Copy-Move forgery

(a) (b)

Fig. 2(b) Image splicing

Fig. 2(c) Image Morphing

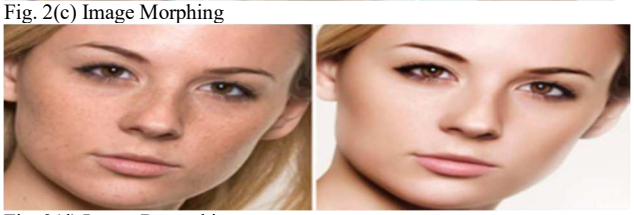

Fig. 2(d) Image Retouching

Fig. 2(c) Image Enhancement

Fig. 2 An illustration of the different types of image forgery cases

Retouching tampering signifies the improvement in spatial and textural features, which is clearly demonstrated in Fig. 2(d). The Fig. 2(e) shows the example of image enhancement. In these images, the upper left corner image shows original image, which is improved and depicted in right top corner. Similarly, the lower left corner image shows the blurred image, which is improved in the right bottom corner image.

Based on the (image) significances and allied criticality ensuring authenticity and tamper-proofing is inevitable. Though, ensuring aforesaid tamper-proof (image) data over gigantic open platforms where the advanced software can be used to tamper digital images in such manner that a normal naked eye can't detect any manipulation, is really a challenge [1]. The frequency of the digital image tampering or forgery is on rise, which has been alarming academia-industries to develop more efficient and robust image forgery detection tool, which could detect, locate and classify forged images for communication and analytics. In sync with aforesaid challenges and allied scopes, different image forgery detection and classification systems are developed [2].

In contrast to traditional methods for detecting image forgery, deep learning-based techniques have shown superior performance in the detection and classification of forgeries. In the past, the authors have applied stereotype methods like block-based methods and key-point driven solutions. To accomplish this, the authors employed diverse manually designed features such as discrete cosine transformation (DCT), discrete wavelet transformation (DWT), principal component analysis (PCA), spatially invariant Fourier transformation (SIFT), SURF methods, and others. Despite numerous efforts the majority of these solutions are unable to yield accurate forgery detection. Though, deep learning driven solutions [14] have performed superior over aforesaid methods.

The deep learning-based methods using the hierarchical features encompassing low-level, mid-level and high-level features to perform forgery detection and classification [3]. To achieve better performance, a few deep models have been proposed [3] where at first input images are split into multiple patches, where from each patch it extracts features by using stacked auto-encoder to perform feature learning and classification. Though, a few methods make use of the contextual features to achieve higher accuracy. More innovative efforts have been made in the past, where more advanced deep models like VGG-16, improved CNNs driven solutions [5], are developed. However, generalizing their efficacy towards real-world image forgery detection seems to be a challenge. In this paper, the different state-of-arts pertaining to the image forgery detections are discussed. More specifically, this study explores deep learning techniques for detecting copy-move forgery and identifying image splicing attacks etc. This paper discusses various deep learning models such as CNNs, DNNs, ResNet, and others that are developed for detecting image forgery. The different state-of-arts for their strengths as well as limitations are also discussed in this paper.

II. Literature Survey

This section addresses some of the key literature related to the deep learning-based image forgery detection systems. More specifically, the deep learning models designed towards copy-move forgery detection, splicing attack detection, resampling manipulations etc. are discussed for their strengths as well as limitations.

A. Copy-Move Forgery Detection

Out of various types of image tampering cases, copy-move forgery is the most prevalent.. Figure 3 provides an example of copy-move forgery cases.

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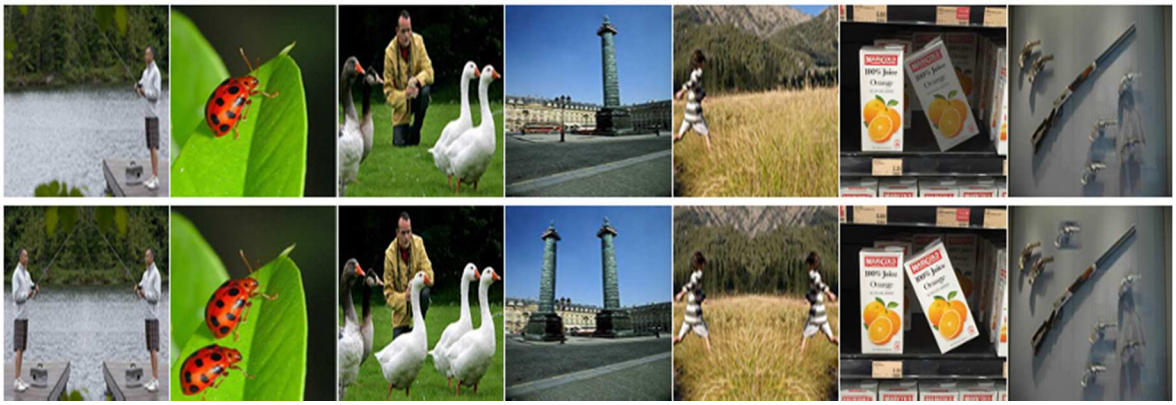

Fig. 3 Visualization of various instances of copy-move forgery

The traditional copy-move forgery detection methods, especially applying hard-crafted features [6] often result unsatisfactory outputs. These traditional methods primarily make use of three phases, in which at first each phase is trained distinctly where it demands manual tuning of the numerous parameters [6-8]. Subsequently, these methods require tuning to achieve superior performance over the different datasets [6]. Finally, the handcrafted features are typically have limited or confined discrimination ability [8][9]. Unlike conventional approaches, copy-move forgery detection methods utilize deep learning techniques [10]. Despite the fact that selecting the suitable parameters and allied threshold confine their robustness over the different copy-move forgery datasets [7]; yet, they possess great ability to learn over complex and non-linear features to perform copy-move forgery detection and classification systems [7]. Though, the recently proposed transfer learning methods, especially applied over each block [26] and key-points can result superior efficacy towards copy-move forgery detection [25]. The key transfer learning methods like AlexNet, VGGNet, GoogleNet, and ResNet have been applied to train over significantly large training dataset and hence possesses a vital generalization ability [12].

The recently developed copy-move forgery detection methods [8][9] made use of the deep learning frameworks, where the deep models are applied on patch-wise processing. In these methods, the deep learning models extract features from each block [26] or key-points to perform copy-move forgery detection [15]. The authors in [13] applied AlexNet deep models over each image block to perform forgery detection and classification. Similarly, a general processing unit (GPU)-based convolutional kernel network (CKN) [8] driven local descriptors are designed to extract key-points for learning and prediction [25]. In this method, CKM was designed as a deep convolutional network by amalgamating different neural networks with the distinct kernel techniques that eventually enabled generation of the local feature descriptors which were static over the different transformations. Moreover, the authors performed segmentation followed by CNN-based feature extraction to improve learning outputs. Interestingly, the majority of the transfer learning driven solutions have performed image-level forgery detection and classification. However, their scope is merely as feature extraction tool and hence eventual efficiency over the different classifiers remained an open research area. A number of state-of-arts have used synthesized dataset by applying a large number of images containing forgery or manipulation localization binary mask to improve training efficiency for copy-move forgery detection [9][14].

In real-world applications, the forged images often depict exceedingly high similarity amongst the genuine and forged regions, and therefore, merely applying CNN driven local features can't yield generalizable performance. To tackle this issue, the authors [14][10] introduced a hybrid deep model combining CNN and long short-term memory (LSTM) architectures. This model aims to identify, locate, and classify various types of image forgeries, including copy-move, image splicing, resampling, and removal. The authors [10][14] applied LSTM to address long-term dependency problem while retaining contextual details. This model facilitated learning the differences between forged and original regions, leading to enhanced accuracy in detection. The authors [23] designed LSTM with five convolutional layer which helped to retain high-dimensional features for further learning and classification. In this method, initial two layers processed over input image patches that yielded low-level feature map to perform learning and prediction. The authors applied the subsequent three convolutional later to retain the information rich feature map for learning and prediction. In reference [16], the authors initially utilized a Siamese network to distinguish between different classes by learning intrinsic features through a distance metric. In their deployed network the Siamese network encompassing twin sub-network which helped to classify whether the two images are same as not [16]. The authors [16] applied Siamese network to detect hidden forgery regions or traces. Additionally, they applied a forgery localization technique [16] by splitting an image into overlapping patches which were subsequently compared by means of the Siamese network to confirm whether those sections are similarly processed or not. In case the regions are processed differently, it was annotated as forged. The authors [17] developed a Faster R-CNN deep model incorporating two parallel streams: RGB and noise streams, to detect image forgery. In this approach, the RGB stream identifies the overall visual tampering artifacts. The spatial rich model (SRM) was used to capture local noise features, aiding in the evaluation of noise inconsistencies. Ultimately, a bilinear pooling layer was employed to integrate the aforementioned RGB and noise streams, enhancing the learning and prediction process. The authors [9] applied Mask R-CNN

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model to perform forgery detection and classification. The authors applied a Sobel-based edge detection, which was followed by Mask R-CNN deep model for feature extraction and learning towards image forgery detection and classification. The authors [18] proposed a dense UNet segmentation model, which was followed by SRM filters to learn and predict image forgery. The authors conducted multi-scale up-sampling, followed by concatenating convolutional features, to leverage contextual or global features for detecting and localizing forgeries. The deep neural network model was applied to detect and localize splicing forgery. In this method the authors used tampering traces [16] to perform forgery detection.

Luo et al. [19] developed a dual attention forgery detection network (DAFDN) that combines a spatial reduction attention block (SRAB) with a forgery feature attention module (FFAM). DAFDN resembles CNN in function and detected strange traces by learning spatial details. It could achieve the average area under curve as 0.925. Li et al. [20] devised an artifacts-disentangled adversarial learning (ADAL) model designed to detect deepfakes. ADAL operated as a multi-scale feature separator (MFS), enhancing the transmission of object features via optimized encoder-decoder connections. Furthermore, an artifacts cycle consistency loss (ACCL) function was developed to utilize disentangled artifacts in generating new samples, thereby enabling pixel-level supervised training for detecting manipulations. It also applied the parallel symmetric discriminators to classify original and the fake images.

Li et al. [21] too designed an attentional cross-domain deep model for image manipulation detection. It used a discriminative feature layer where it fused the rich features obtained from the different domains to yield a trainable feature vector to enable accurate image tampering detection. They proposed tampering discriminative attention network (TDA-Net), which helped to focus on the suspicious elements within an image. Chen et al. [22] designed an anti-forensic attack model, where attack angle and scale and mutual information scale features were applied to represent the forensic feature variation for tampering detection. Yang et al. [23] exploited pixel-wise spatial and temporal signatures to detect deepfake tampering. Thay also proposed spatio-temporal faint synthetic signals embedded in the malicious videos to detect deepfake tampering. They used ACBlock-based DenseNet for deepfake prediction.

B. Splicing Forgery Detection

Unlike copy-move forgery detection methods, the forger prepares composite images in such manner that it becomes difficult to identify the authentic or forged images. It is really a challenging task while looking at an image at the first glance. Similar to the copy-move forgery cases, numerous efforts have been made towards image splicing manipulation detection. In general, splicing refers the process of combining two or multiple images. Unlike copy-move forgery cases, image splicing doesn't possess image duplication and therefore makes (forgery) detection a challenging task. The key characteristics of the image splicing is based on the clues which are left post tampering. For instance, the discontinuity in edges, pixel's inconsistency, geometric changes and camera conditions provide certain clues to detect image splicing. Despite this, due to the image splicing, some image specific statistics gets disarranged; though remains imperceptible by naked eyes. The artificial intelligence methods exploit such disorientation or disturbances to detect splicing forgery in the digital images [41][42].

Shi et al. [43] utilized multi-size block discrete cosine transform (MBDCT), incorporating statistical features such as moments of characteristic functions of wavelet sub-bands and Markov transition probabilities from input images, for splicing forgery detection. Dong et al. [44] developed a technique for detecting image splicing using statistical features derived from image run-length representation and image edge statistics. They employed a support vector machine (SVM) as a classifier, achieving a detection accuracy of 84.36%. Wang et al. [45] introduced a novel technique utilizing the image chroma component. Hsu et al. [46] employed a fully automatic approach for detecting splicing forgery in digital images, incorporating three distinct features: geometry-invariant CRF estimation, consistency checking, and image segmentation. Even with such extensive computation, this method achieved a recall and precision rate of 70%. Kakar et al. [47] introduced a method for detecting image forgery that leverages discrepancies in motion, such as blur and spectral analysis of image gradients. Carvalho et al. [48] developed color's inconsistencies across images that in conjunction with the SVM classifier performed image splicing forgery detection. Rao et al. [49] proposed splicing forgery by exploiting motion blur features. Their simulation results over CASIA1 and CASIA2 datasets showed the prediction accuracy of 95.6%.

C. Resampling Forgery Detection

Image resampling represents a mathematical mechanism to alter image's resolution and their volumetric quantity (say, samples), primarily intended to increase sample size. Such methods are applied especially for banner's printing and hoardings, etc. It is also done to reduce image samples by performing down sampling for web-application like email and website purposes. In general, it includes various types of forgery cases that involve scaling, rotation, or skewing operations to manipulate images. In such functions resampling and interpolation mechanisms turns out to be decisive, and therefore it becomes feasible to detect the forgery or image manipulation by exploring the characteristics of image's resampling. Popescu et al. [50] detect digital image forgeries by detecting resampling forgeries. To detect resampling traces, interpolation information can be a fundamental telltale feature to detect resampling forgery. The authors [51] made use of expectation maximization model to measure whether a sample has been resampled. In this reference, the authors developed two distinct solutions, first those samples which are correlated with the neighbors, while the second method represents those images which are not correlated. While this method can detect linear or cubic interpolation, it struggles to identify other advanced nonlinear interpolation methods.

III. Datasets for Image Forgery Detection

There are very small datasets available towards image forgery detection and classification. The depth quantification of the

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literatures has helped us to identify certain publicly available datasets, especially prepared towards image forgery detection, especially for the copy-move forgery detection, splicing forgery detection, resampling, retouching. A snippet of the key datasets is given in Table I.

Table I: The Datasets available towards image forgery detection and classification

Dataset	Forgery type	Total images	Resolution	Data Distributions
CISDE [28]	Splicing	1845	128 ×128	There are a total of 933 forged images and 912 authentic images available in grayscale format (*.PNG).
CUISDE [29]	Splicing	361	757 ×568, 1152 × 768	It contains 180 forged images and 181 normal images. These images are available in *.TIFF format.
CASIAv1.0 [30]	Splicing	1725	324 ×256	It embodies a total of 925 forged images, along with the 800 normal or authentic images. These datasets are available in *JPEG format.
CASIA v2.0 [31]	Splicing	12614	240 × 160–900 × 600	This dataset includes 5123 forged images and 7491 authentic images, all stored in *.JPEG format. It contains both uncompressed images and JPEG images with various Q factors.
CMFDA [32]	Copy-move	48	420 × 300–3888 × 2592	It embodies authentic as well as forged image, which are stored in *.JPEG compression, format.
CoMoFoD dataset [33]	Copy-move	260	512 × 512–3000 × 2000	This data contained original and forged images, which were obtained by performing translation, rotation, scale, distortion.
MICC-F600 [34]	Copy-move	600	800 × 533–3888 × 2592	It included both genuine and manipulated images randomly selected from the MICC-F2000 and SATS-130 datasets.
MICC-F2000 [35]	Copy-move	2000	2018 × 1536	It embodies the authentic and forged images, which are obtained by performing translation, rotation, scale.
SBU-CM161 [36]	Copy-move	240	800 ×580	It consists of images created from 16 original JPEG images that have been manipulated with rotation, scaling, and compression.
CPH [37]	Copy-move	216	845 × 634–296 × 972	It includes images containing forgeries produced through a combination of operations such as resizing, rotation, scaling, compression, and illumination matching.

IV. DISCUSSIONS

The overall discussion of the image tampering detection and analysis model revealed that the processes of image tampering detection predominantly emphasize on the coarse-grained analysis, while forgery localization requires fine-grained assessment. In this manner, locating certain manipulation or tampering within an image is a challenging task. The traditional block and key-point based methods often makes of the hand-crafted features to detect image tampering. However, such approaches can't yield generalizable performance over the complex and non-linear forgery components or an image with complex structure and objects. Unlike traditional approaches, the deep learning methods exploit abstract and complex deep features to detect and identify the forgery. The majority of the deep learning models exploit spatio-textural deep features to perform binary classification to detect and classify authentic images and forged images. Though, CNN-based methods have achieved higher accuracy for both forgery detection, classification as well as localization. Here, the fine-grained mask obtained by the CNN models helps to localize the forged regions. Yet, the complete reliance onto the local hierarchical features confines their robustness, especially over large complex images. Moreover, classifying the different types of image manipulation, especially when the user is not known of the manipulation types, is really a challenge with CNN-based solutions.

The study revealed that the deep learning-based models demand large number of annotated or labelled data (i.e., digital images), which can be difficult over the real-time data conditions. It forces the state-of-arts to prepare data and feature models specific towards a target forgery type. This reference introduces various deep learning methods such as CNN, LSTM, among others. Though, the efforts are also made by designing improved deep networks like AlexNet, ResNet, MobileNet etc. to enable more accurate forgery detection and prediction. Unlike the traditional block-based or key-frame based measures which often use hand-crafted features for forgery detection, the deep learning-based methods are more efficient, time-efficient and accurate. Deep learning-based methods typically involve three main stages:

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feature extraction, feature matching, and forgery detection and localization. However, these methods often lack certain abilities or suffer from challenges such as the gradient vanishing, lack of contextual details, long-term dependency and accuracy degradation over large convolutions. To improve efficiency, the deep models require improve feature extraction by amalgamating features from the small, smoothed patches and low-resolution images. It might also demand certain geometric transformation and allied post processing tasks to enable higher detection accuracy.

The use of distributed key-points detection followed by feature extraction can yield higher efficacy, even over low-resolution images. The use of hybrid deep features too can improve features by retaining local as well as contextual information that consequently can yield higher accuracy. The literatures also reveal that the complexity involved in feature matching is also a key challenge towards copy-move forgery detection. In addition, increasing the extracted features, especially over small/smooth cloned regions impose great challenge in matching based copy-move forgery detection and classification. On the other hand, using low-dimensional feature descriptors can decrease computation time; yet, can't support generalizability due to the compromised performance. To address such challenges performing feature reduction with certain optimally designed feature selection methods and exploiting most significant features can yield higher accuracy even at the reduced computational costs and time. For localization of forged regions, one should use optimally designed descriptors possessing high discrimination ability, and suitable threshold values, etc. The use of region of interest (ROI) segmentation too can help reducing computational cost, while addressing class-imbalance problem. The deep features can further be improved by amalgamating attention layer which can yield more discriminability to make learning superior and accurate towards image forgery detection and classification.

The above discussions infer that in the past numerous efforts have been made towards image forgery detection; yet, generalizing the efficacy over the non-linear and ambiguous images remains a challenge. Moreover, lower accuracy and at hand deep related challenges such as the gradient vanishing, gradient explode, accuracy degradation and long-term dependency limit the efficacy of the major state-of-arts. It broadens horizon for the other researchers to design and develop a more efficient and robust image forgery detection solution, which can be suitable to detect the different kinds of attacks without demanding large annotated samples for training. It can be considered as the motivation for future efforts.

V. Conclusion

In this paper, the different deep learning-based image forgery detection and classification systems were discussed. Various deep learning techniques such as CNNs, RNNs, AEs, and advanced transfer learning methods aimed at detecting copy-move forgery, splicing, resampling, and other forms of manipulation are discussed, including their strengths and limitations. This survey revealed that in the past numerous efforts have been made towards image forgery detection; yet, generalizing the efficacy over the non-linear and ambiguous images remains a challenge. Moreover, lower accuracy and at hand deep related challenges such as the gradient vanishing, gradient explode, accuracy degradation and long-term dependency limit the efficacy of the major state-of-arts.

This survey also indicated that unlike traditional deep learning-based solutions where standalone local features are used to perform forgery detection and classification, the use of hybrid deep models, strategically designed to exploit both local as well as contextual information can be of paramount significance.

VI. Future Scope

This survey broadens the horizon for the other researchers to design and develop a more efficient and robust image forgery detection solution, which can be suitable to detect the different kinds of attacks without demanding large annotated samples for training. It can be considered as the motivation for future efforts. The deep features can also be amalgamated with the spatio-textural features which can help with the different statistical distribution details to improve learning and classification results. Such methods can be suitable for the different image forgery detection and classification purposes. Though, there exists a number of datasets towards image forgery detection and classification; however, the likelihood of data imbalances, lack of annotations, ambiguities over run-time environment etc. can't be ruled out. Rather processing the complete original images as input, the authors might also explore the efficacy of semantic segmentation methods, which in conjunction with the deep features can not only alleviate class-imbalance, but can also improve prediction accuracy over run-time environment. These key inferences can be considered as the scope for further innovations.

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